

The Spatial Performance Assessment for Cognitive Evaluation (SPACE): A Novel Game for the Early Detection of Cognitive Impairment

Giorgio Colombo*
giorgio.colombo@sec.ethz.ch
Singapore-ETH Centre
Future Health Technologies Programme
Singapore

Jascha Grübel*
jgruebel@ethz.ch
ETH Zürich
Chair of Cognitive Science
Game Technology Center
Switzerland



Figure 1: SPACE. A novel game for the early detection of cognitive impairment.

ABSTRACT

Dementia is a leading cause of dependency and death among older adults and an economic burden to healthcare. In the fast-paced non-gaming environment of healthcare, the detection of dementia is limited to invasive and expensive procedures (e.g., PET) or less sensitive paper and pencil tests. Typically, dementia is preceded by Mild Cognitive Impairment (MCI) which is characterised by the loss of cognitive function including deficits in spatial ability. Spatial ability may be a sensitive marker for discriminating between healthy individuals and MCI patients. Virtual Reality now allows for spatial tests to be easily and safely deployed in clinics or at home. We introduce the Spatial Performance Assessment for Cognitive Evaluation (SPACE) as a transgressive, transformative gamified battery of spatial tests for tablets. SPACE is designed to identify

differences in spatial ability to support clinicians in the early detection of dementia and offers a flexible ecosystem for immediate and longitudinal assessments.

CCS CONCEPTS

• **Computing methodologies** → **Virtual reality**; • **Human-centered computing** → **Empirical studies in interaction design**; **Empirical studies in HCI**; • **Applied computing** → **Consumer health**; **Health informatics**; • **Social and professional topics** → **Seniors**; • **Software and its engineering** → **Interactive games**.

KEYWORDS

Mild Cognitive Impairment, Health Game, Spatial Ability, Dementia, Cognition

*Both authors contributed equally to this research.

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1 INTRODUCTION

Dementia is a neurodegenerative syndrome of cognitive functions that affects approximately 55 million people worldwide. Dementia cases are expected to rise to 139 million in 2050, and the costs associated with it are projected to surpass US\$ 2.8 trillion by 2030. Typically, dementia is preceded by a transitional stage known as MCI in which cognitive deficits are mild and spatial ability is compromised [28]. Indeed, research has shown that individual differences in navigation ability may constitute an important marker for discriminating between different levels of cognitive impairment [6]. For example, Howett et al. [9] have shown that a Virtual Reality (VR) navigation task (i.e., path integration) can differentiate between patients with mild cognitive impairment at low and high risk of transitioning to dementia more accurately than gold standard paper and pencil tests. Despite these findings, the use of navigation tasks for the detection of cognitive impairment has not successfully translated to clinical settings. This shortcoming may be related to the usability of the VR tasks, the limited sample size employed in previous studies, and the time required to complete the assessment.

SPACE was developed to bridge the gap between research and clinical application by developing a cost-effective and easy-to-administer assessment to investigate the link between individual differences in navigation ability and cognitive impairment (see Fig. 1). SPACE is a flexible tool that allows researchers and clinicians to design and administer personalised modular tasks in a gamified environment. The engaging gameplay moves beyond rudimentary laboratory experimental designs and opens the door for quick deployment with large samples in clinical settings and at home. Here, the ability to deploy SPACE at home can reduce the number of unnecessary diagnostic procedures while potentially identifying individuals at an early stage of the disease that might need follow-up clinical assessments. In addition, the SPACE ecosystem offers the possibility for users to easily export the collected data in an organised JSON file for analysis. Through a variety of different configuration panels and data loading and extraction functionalities, SPACE adheres to the Design, Experiment, Analyse, and Reproduce (DEAR) principle [7] and overcomes previous challenges related to the generalisability and reproducibility of results. As such, SPACE is a *mobile laboratory* that provides healthcare professionals, without any prior knowledge of coding, the flexibility to configure and run state-of-the-art spatial tests.

2 BACKGROUND & RELATED WORK

MCI and Alzheimer's Disease (AD) are known to adversely affect human hippocampus and entorhinal cortex through extracellular plaque deposits of the β -amyloid ($A\beta$) peptide [1] and neurofibrillary tangles of the microtubule-binding protein tau [10]. These two regions also play a critical role in the spatial coding of locations [18] and in assisting individuals to keep track of changes in their position and orientation as they move through an environment [8, 20]. Indeed, patients diagnosed with MCI and AD, experience a decline in cognitive function (e.g., attention and memory) in addition to deficits in spatial and navigational ability [12]. Here, atrophy and reduced neural activity in the hippocampus and entorhinal cortex have consistently been associated with reduced performance in a

variety of spatial tasks in patients with MCI and AD compared to healthy controls [13].

Interestingly, the most common paper and pencil tests (e.g., Montreal Cognitive Assessment, MoCA [16]) and apps (e.g., Altoida [14]) used to detect cognitive impairment do not focus on spatial and navigation tasks. Researchers have now started to use VR and gamification to probe the relationship between spatial navigation deficits and cognitive impairment [9]. There are several advantages to using VR in navigation research, particularly when working with the elderly. First, VR provides safer and more controlled environments for navigation studies that require the strict manipulation of specific variables (e.g., landmarks, visibility). Second, VR allows for the accurate collection of movement (e.g., rotation, translations) and interaction (e.g., pressure) data. Third, VR can be used to access hard-to-reach populations (e.g., elderly, disabled) often excluded from research. Fourth, VR is cost-effective, efficient, and allows individuals to mimic real-world actions in a scaled environment with limited impact on ecological validity [5]. Here, techniques such as teleportation reduce delays between trials when participants are required to move between locations in the environment. Fifth, experiments in VR are easier to replicate and share between professionals, allowing for better validation of results and monitoring of progress. Sixth, VR can support engagement, increase compliance and reduce attrition arising from difficult and repetitive tasks [2].

Despite these advances, there are several challenges to conducting navigation research in VR. VR experiments can be costly to develop and some testing paradigms have been criticised for their limited realism and inability to capture the complexity and nuances (e.g., proprioceptive feedback) of real-world navigation. Some participants also experience simulator sickness when engaged in virtual tasks while others, particularly the elderly, have considerable difficulty interacting with the controls (e.g., touch and swipe). VR experiments have also been criticised for their limited generalisability to real-world applications. Indeed, a recent publication suggests that participants commit a systematic perceptual bias in a test that required them to estimate the position of objects after shifting perspectives in VR [21].

The success of gamified assessments for detecting cognitive impairment has been mixed [14, 24]. While some assessments that include spatial tests have shown promising results in their ability to identify age-related decline across the lifespan [25], the sensitivity to detect differences between MCI, AD and healthy participants remains to be determined given the limited sample of these studies [9, 19]. One notable example of a gamified assessment that was designed to investigate the relationship between navigation deficits and cognitive impairment is Sea Hero Quest (SHQ) [15]. In SHQ, participants are asked to captain a ship through a variety of environments and complete a series of spatial tasks that test their capacity to build a cognitive map of these environments. Despite the high level of reported engagement and widespread distribution (over 4.5 million players worldwide), the ability of SHQ to detect early signs of cognitive impairment has been limited to laboratory findings on individual differences in navigation (e.g., gender, economic disparity, the entropy of city networks) and the genetic predisposition of individuals to the disease [3, 4]. This difficulty in linking deficits in

navigation ability to cognitive status may be related to the simplification of the spatial tasks, the lack of controlled testing and the challenges associated with interacting with the controls.

SPACE attempts to address many of these shortcomings by implementing more precise measures of spatial ability and introduces simplified controls to facilitate game interaction when testing elderly groups. In addition, the SPACE ecosystem provides researchers and clinicians with a variety of menus to configure and adapt their game sessions. Currently, the capacity of SPACE to predict cognitive status is being validated in large cohorts ($n = 1500$) of participants in controlled clinical settings. If successful, SPACE has the potential to revolutionise healthcare programs by providing personalised assessments leading to a patient-centred approach that can lead to more targeted diagnostics and treatments.

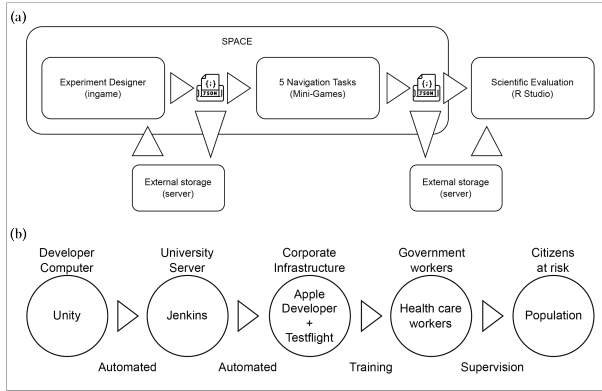


Figure 2: Architecture and Development of SPACE. 2a) The application of the DEAR principle in SPACE. 2b) The production and distribution of SPACE.

3 DESIGN & DEVELOPMENT

We created SPACE to be both engaging and scientifically rigorous. The game is set on a foreign planet because it provides a visually appealing backdrop and a challenging navigation environment void of salient visual cues that is essential for testing the acquisition of spatial knowledge. In SPACE, players assume the role of an astronaut who has been sent on a mission to uncover the potential of a new planet to harbour life. Following an extensive training phase, players are asked to complete five spatial tasks designed to engage different aspects of their navigation ability. SPACE collects data on position and orientation, time, coordinates, and screen interaction (e.g., dragging time and pressure) and produces a JSON file with scores for each task. Individual or aggregated scores are used to isolate differences in spatial ability that may constitute markers of cognitive status.

SPACE is implemented in Unity3D and uses JSON files to exchange information on the experimental design and participant data with trusted remote servers. The architecture is split into three parts (see Fig 2a), in line with the DEAR principle. The Experiment Designer allows researchers to design their own experiments, the 5 Navigation Tasks define the experiment, and the Scientific Evaluation constitutes the analysis of the collected data in R. Reproducibility and validation are made possible by storing intermediate

data as JSON files remotely, sharing experiment designs, and safely storing user data. Game production is supported by a GitLab LFS-repository automatically built on a Jenkins server and deployed to iOS via TestFlight making it possible for deployment in the population (see Fig 2b).

The game has been optimised for elderly citizens and was designed to be mostly self-explanatory. To achieve this, we have developed various types of controls (i.e., Tap & Swipe, Joystick, and Anchor) that cater to different experience levels and may assist in overcoming difficulties when interacting with tablets. We have also designed a companion robot that guides players through various spatial tasks. Specifically, the robot is scripted to provide players with instructions on the tasks, give encouraging gestures during gameplay and feedback on task performance. In addition, players experience the game through the viewpoint of an astronaut's helmet implemented as a themed User Interface (UI) to assist with immersion. To further support the administration and deployment of SPACE in a variety of settings, we have also created a series of instructional videos that can be presented ahead of each task.

3.1 Gameplay

SPACE includes a training phase and five spatial tasks (i.e., path integration, egocentric pointing, mapping, associative memory, and perspective-taking). At the end of each task, players receive feedback and are awarded a different colour badge (i.e., bronze, silver, gold) according to their performance.

Training. Researchers have found that ability to interact with the controls can be a confounding factor when assessing navigation performance (see Fig. 3). SPACE starts with a comprehensive training phase in which players are tested on basic visuospatial skills while they learn the different controls that will be used in the game. Specifically, players learn to rotate and translate and to integrate these movements by following the robot in a circuit around the

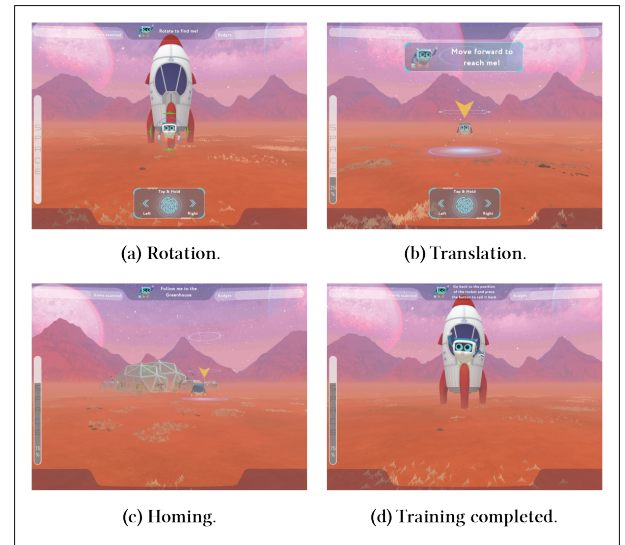


Figure 3: Training. Players practice with the various controls by completing a series of manoeuvres around the planet.

planet. During training, players are also introduced to the logic of the first task (i.e., path integration) to minimise learning effects during testing.

Path integration. In this task, players start their journey at the rocket and are asked to follow the robot to two distinct landmarks in the environment before returning unguided back to the rocket (see Fig. 4a, 4b). At each landmark, the robot scans an item that players will be asked to recall, along with the location of the landmarks, in subsequent tasks in the game.

Egocentric pointing. In this task, players test their memory for the locations encountered during the path integration task by pointing between landmarks (see Fig. 4c).

Mapping. In this task, players test their configurational knowledge of the environment. Specifically, they will be asked to create a map of the environment by dragging and dropping six icons representing the locations they learned during the path integration task. After dragging all the icons, an animation with the correct position of the landmarks is displayed (see Fig. 4d).

Associative memory. In this task, players are asked to pair the items scanned by the robot with the corresponding landmarks they encountered during the path integration task. After participants complete the task by dragging the icon of the element on top of the corresponding icon of the landmark, an animation is triggered showing the corrected pairings (see Fig. 4e).

Perspective-taking. In this task, players are asked to imagine standing at a landmark facing another landmark by looking at the corrected map of the environment. Players are then required to indicate the correct bearing towards a third landmark (see Fig. 4f).

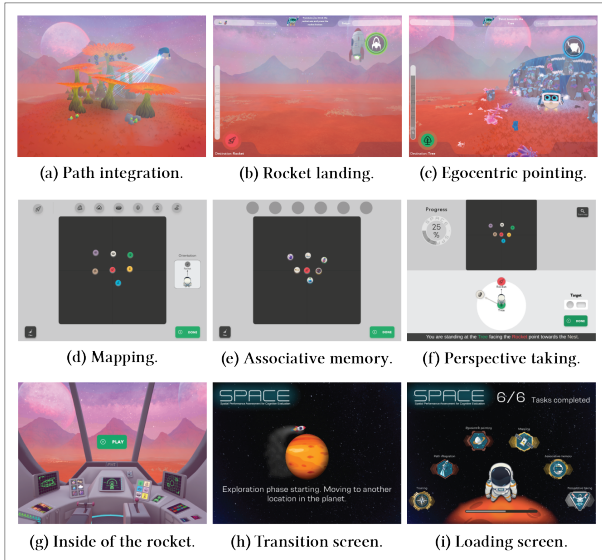


Figure 4: SPACE tasks. Screenshots of the five tasks in SPACE and loading screens.

We selected these five spatial tasks because previous research has validated their capacity to distinguish between good and bad navigators [26] which may underlie pathological changes in the ageing brain [9]. Specifically, path integration relies on grid cells in the

entorhinal cortex that allow humans to keep track of their position and orientation as they navigate through the environment [8, 22]. This task has been capable to differentiate between healthy ageing and cognitive impairment and classify different MCI subgroups better than other neuropsychological tests [9]. Similarly, success in the egocentric pointing and mapping tasks relies on the healthy functioning of place cells in the hippocampus that are responsible for accurate location encoding [18]. Along with the perspective-taking task, these tasks will allow us to probe the successful use of different spatial strategies (egocentric vs. allocentric) that have been found to further differentiate cognitive functioning [23].

3.2 SPACE ecosystem

SPACE has been designed to allow researchers and clinicians, even those with limited coding ability, to easily configure each of the tasks according to their needs. Through various configuration menus (see Fig. 5a, 5b, 5c), users can set the number and position of landmarks that are included in the configuration along with the order and characteristics of each trial. Inherent to the menus, is a logic system allowing experimenters to proceed only if all the trials generated are possible to be executed in the virtual environment. Other functions such as repetitions, self-generation of trials and randomisation offer novel and quick ways to design and conduct experiments on the fly. Finally, users can verify, in real-time, measures of the angular and distance relationships between landmarks to define different levels of difficulty.

Using the Experimenter menu (see Fig. 5d), users can select the language, type of control, the speed of the robot and astronaut, set the volume of sound effects and music, and decide whether to include up to three pre- and post-game questionnaires. Users can also decide which spatial tasks (including training) will be part of the assessment along with the presentation order and session repetitions. SPACE was designed to facilitate deployment in clinical

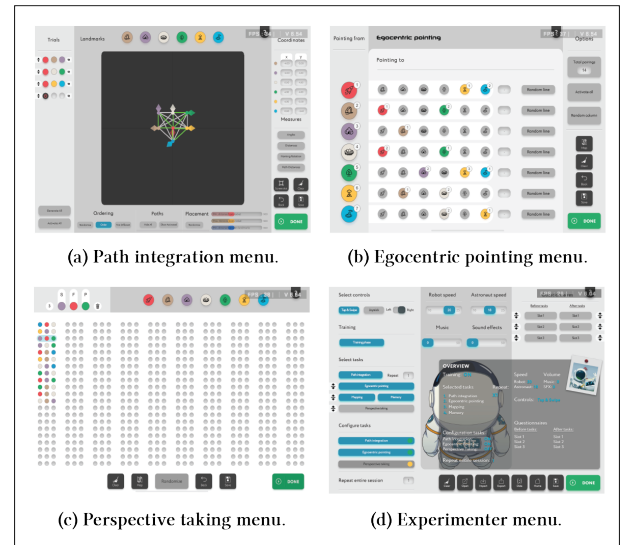


Figure 5: Menus in SPACE. Screenshots of the four configuration panels in SPACE.

settings where time is of the essence and the VR experience of the clinician is limited. Through shortcut buttons or by importing JSON files in the Experimenter menu, clinicians can quickly load, administer and share predefined configurations for their assessments in a consistent manner across clinics worldwide.

4 CRITICAL REFLECTION & CONCLUSION

The World Health Organisation (WHO) estimates that there are nearly 10 million new dementia cases each year [28]. Caring for patients with dementia is associated with a variety of physical, psychological, social and economic repercussions that can no longer be supported by our current healthcare system [27]. To date, the diagnosis of MCI is relatively complex and there is no single test that can reliably detect it. In fact, clinicians must perform several assessments based on patients' symptomatology and perform several clinical (e.g., medical history, memory assessments, and paper and pencil tests), neurological (e.g., eye movement and gait), fluid (e.g., blood tests) and radiological (e.g., MRI) biomarker tests to make an accurate diagnosis [11]. Recently, researchers have shown that VR can be used in the assessment of cognitive status since it is capable of capturing many of the cognitive processes (including spatial ability) that are compromised by pathological ageing [5, 14]. However, the deployment of digital assessments in clinical settings has faced a series of challenges relating to the sensitivity and specificity of the tool, the usability, the cost-effectiveness, and the reproducibility of findings. SPACE attempts to address these shortcomings by providing a gamified assessment capable of identifying spatial deficits that may serve as a digital biomarker to detect early signs of cognitive impairment.

Despite their widespread use in clinical settings, paper and pencil tests tend to focus on memory and language and typically overlook the role of spatial abilities. In addition to requiring trained personnel for their administration, these tests have also been found to be susceptible to educational and cultural biases and lack the necessary sensitivity to effectively differentiate between varying degrees of cognitive impairment [17]. In contrast, SPACE was specifically designed to tap into the neurological processes that link spatial deficits to cognitive impairment. The different spatial assessments in SPACE test whether the place and grid cells in the hippocampus and entorhinal cortex have been compromised by pathological ageing. Existing gamified assessments such as SHQ[3] and the *Leiden Navigatie Test* [25] that test spatial ability have used oversimplified versions of these tasks that reduced the sensitivity to detect differences in cognitive impairment.

SPACE attempts to balance the need for user engagement inherent in gamified assessments while employing tasks with the complexity necessary for accurate diagnosis. From the beginning, SPACE was designed to cater to the needs and technological skills of an ageing population. Towards this end, we set up a focus group with elderly individuals and conducted a series of pilot usability studies to identify and improve various aspects of the game. Results from these studies revealed that elderly participants struggled to use the control interfaces (Tap & Swipe and Joystick), to keep track of the robot on the planet, read text instructions and understand some of the logic of the spatial tasks. To address these issues, we implemented a new simplified "Anchor" control that constrains interactions and allows players to navigate by pressing and holding

a button. We have also developed a series of UI features including zooming in-game instructions, call-out bubbles and shaders that highlight the robot's position when it is at distance from the player, and progress bars or loading and transition screens that update as the player advances through the various trials and tasks (see Fig. 4h, 4i). Finally, for easier administration in clinics and laboratories, we translated the game into eleven different languages and created a series of instructional videos that are played at the beginning of each task. These videos provide players with a short overview of the task requirements in order to reduce confusion and minimise learning effects.

Healthcare is often regarded as a non-gaming environment where time and money are precious commodities. SPACE attempts to modernise and transform the way clinical cognitive assessments are conducted by providing a tool that can be safely administered at the clinic or independently at home. Embedded in SPACE, is an ecosystem of menus that allows users to upload, set up and share predefined configurations of assessments. This ecosystem helps SPACE to move beyond laboratory testing and towards reproducible citizen science whereby individuals, researchers and clinicians all work together to collect data in order to develop more sensitive norms for the digital detection of early signs of cognitive impairment. To date, we have collected limited data on SPACE. Nonetheless, our preliminary results are promising and show that the tasks in SPACE are sufficiently robust to capture age-related decline in spatial abilities. We are now validating SPACE relative to gold standard paper and pencil neuropsychological assessments as well as fluid (e.g., blood) and neuroimaging (MRI) biomarkers in large cohorts ($n > 1500$) of participants in Singapore. These efforts to comprehensively validate SPACE will further inform the appropriate implementation and distribution of the game in society while highlighting the potential risks related to the inaccurate detection of the pathology.

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